

Ankur Srivastava

professor

基本信息

姓名: Ankur Srivastava

所属院校: 马里兰大学-大学城

邮箱: ankurs@umd.edu

地区: United States

年龄范围: 50岁以上

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可, 展现了卓越的研究能力和影响力。

学科领域

信息安全

网络空间安全

电子与计算机工程

集成电路设计与集成系统

研究领域

系统安全与漏洞分析

网络安全理论与技术

安全协议与安全认证

数字集成电路设计

EDA工具与电路验证

嵌入式系统与硬件设计

加密与数据保护

密码学与密码分析

计算机体系结构

集成电路设计与制造

+1

个人简介

Ankur Srivastava, a distinguished professor and the seventh director of the Institute for Systems Research at the University of Maryland, holds a joint appointment in the Electrical and Computer Engineering Department. He received his B.Tech from IIT Delhi and his PhD from UCLA, where he was

awarded the Outstanding Dissertation Award in 2002. His research focuses on high-performance, low-power, and secure electronic systems, including applications in computer vision, data centers, and sensor networks. Dr. Srivastava is an IEEE Fellow and has been recognized with numerous awards for his contributions to chip hardware security and academic leadership.

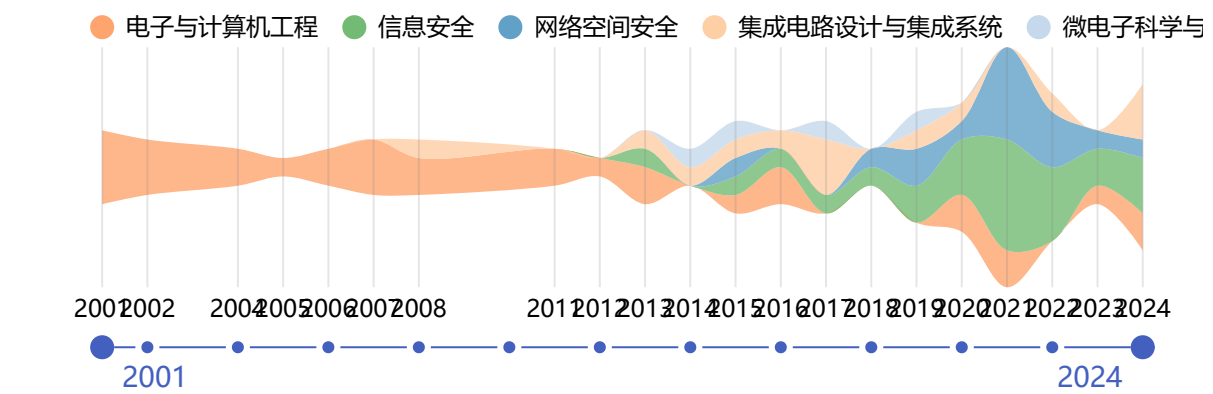
工作经历

University of Maryland - Professor and Director of ISR	2019-07-01 -
University of Maryland - Professor	2014-07-01 -
University of Maryland - Associate Dean	2018-01-01 - 2019-06-30
University System of Maryland - Associate Professor	2009-07-01 - 2014-06-30
University of Maryland - Assistant Professor	2002-10-01 - 2009-06-30

教育经历

University of California - PhD	2000-09-01 - 2002-09-05
Northwestern University - MS	1998-09-01 - 2000-08-01
Indian Institute of Technology - BTech	1994-08-01 - 1998-05-01

近年发表的论文



Removal of SAT-Hard Instances in Logic Obfuscation Through Inference of Functionality
ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS
Mcdaniel, Isaac; Zuzak, Michael; Srivastava, Ankur
被引用次数: 0 2024

A High Level Approach to Co-Designing 3D ICs

Design Automation Conference

Daniel Xing; Ankur Srivastava

被引用次数: 0

2024

Security Evaluation of State Space Obfuscation of Hardware IP through a Red Team-Blue Team Practice

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Rahman, Md Moshir; Geist, Jim; Xing, Daniel; Liu, Yuntao; Srivastava, Ankur; Meade, Travis; Jin, Yier; Bhunia, Swarup

被引用次数: 0

2024

Security Advantages and Challenges of 3D Heterogeneous Integration

COMPUTER

Liu, Yuntao; Xing, Daniel; Mcdaniel, Isaac; Ozbay, Olsan; Akib, Abir; Sukanya, Mumtahina Islam; Rekhi, Sanjay; Srivastava, Ankur

被引用次数: 0

2024

Co-design for Heterogeneous Integration: High Level Decisions to the Rescue

2024 IEEE 67TH INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS, MWSCAS 2024

Xing, Daniel; Akib, Abir; Liu, Yuntao; Srivastava, Ankur

被引用次数: 0

2024

3D-Aware Low Power High-level Resource Binding and Co-Design

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Xing, Daniel H.; Srivastava, Ankur

被引用次数: 0

2024

HQ-DTM: A Hierarchical Q-learning Algorithm for Dynamic Thermal Management of Multi-core Processors

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Akib, Abir Ahsan; Srivastava, Ankur

被引用次数: 0

2024

Introduction to the Special Issue on CAD for Security: Pre-silicon Security Sign-off Solutions Through Design Cycle

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Farahmandi, Farimah; Srivastava, Ankur; Di Natale, Giorgio; Tehranipoor, Mark

被引用次数: 0

2023

Security-Aware Resource Binding to Enhance Logic Obfuscation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zuzak, Michael; Liu, Yuntao; Srivastava, Ankur

被引用次数: 0

2023

Hardware IP Protection against Confidentiality Attacks and Evolving Role of CAD Tool (Invited Paper)

2022 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Bhunia, Swarup; Das, Amitabh; Fazzari, Saverio; Kammler, Vivian; Kehlet, David; Rajendran, Jeyavijayan; Srivastava, Ankur

被引用次数: 0

2022

A Combined Logical and Physical Attack on Logic Obfuscation

2022 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Zuzak, Michael; Liu, Yuntao; McDaniel, Isaac; Srivastava, Ankur

被引用次数: 1

2022

A Black-Box Sensitization Attack on SAT-Hard Instances in Logic Obfuscation

2022 IEEE 40TH INTERNATIONAL CONFERENCE ON COMPUTER DESIGN (ICCD 2022)

McDaniel, Isaac; Zuzak, Michael; Srivastava, Ankur

被引用次数: 1

2022

Evaluating the Security of Logic-Locked Probabilistic Circuits

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zuzak, Michael; Mondal, Ankit; Srivastava, Ankur

被引用次数: 2

2022

Ising-FPGA: A Spintronics-based Reconfigurable Ising Model Solver

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Mondal, Ankit; Srivastava, Ankur

被引用次数: 7

2021

A Resource Binding Approach to Logic Obfuscation

Design Automation Conference

Michael Zuzak; Yuntao Liu; Ankur Srivastava

被引用次数: 0

2021

A Resource Binding Approach to Logic Obfuscation

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Zuzak, Michael; Liu, Yuntao; Srivastava, Ankur

被引用次数: 9

2021

Robust and Attack Resilient Logic Locking with a High Application-Level Impact

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Liu, Yuntao; Zuzak, Michael; Xie, Yang; Chakraborty, Abhishek; Srivastava, Ankur

被引用次数: 9

2021

Invited: Independent Verification and Validation of Security-Aware EDA Tools and IP

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Tan, Benjamin; Garg, Siddharth; Karri, Ramesh; Liu, Yuntao; Zuzak, Michael; Chakraborty, Abhishek; Srivastava, Ankur; Aramoon, Omid; Xu, Qian; Qu, Gang; Porter, Adam; Szep, Jenő; Savage, Warren

被引用次数: 1	2021
Trace Logic Locking: Improving the Parametric Space of Logic Locking	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Zuzak, Michael; Liu, Yuntao; Srivastava, Ankur	
被引用次数: 10	2021
Evaluating the Security of Delay-Locked Circuits	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Chakraborty, Abhishek; Liu, Yuntao; Srivastava, Ankur	
被引用次数: 3	2021
StatSAT: A Boolean Satisfiability based Attack on Logic-Locked Probabilistic Circuits	
Design Automation Conference	
Ankit Mondal; Michael Zuzak; Ankur Srivastava	
被引用次数: 0	2020
Hardware-Assisted Intellectual Property Protection of Deep Learning Models	
Design Automation Conference	
Abhishek Chakraborty; Ankit Mondal; Ankur Srivastava	
被引用次数: 0	2020
Keynote: A Disquisition on Logic Locking	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Chakraborty, Abhishek; Jayasankaran, Nithyashankari Gummidiipoondi; Liu, Yuntao; Rajendran, Jeyavijayan; Sinanoglu, Ozgur; Srivastava, Ankur; Xie, Yang; Yasin, Muhammad; Zuzak, Michael	
被引用次数: 71	2020
Energy-efficient Design of MTJ-based Neural Networks with Stochastic Computing	
ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS	
Mondal, Ankit; Srivastava, Ankur	
被引用次数: 8	2020
DRAMsim3: A Cycle-Accurate, Thermal-Capable DRAM Simulator	
IEEE COMPUTER ARCHITECTURE LETTERS	
Li, Shang; Yang, Zhiyuan; Reddy, Dhiraj; Srivastava, Ankur; Jacob, Bruce	
被引用次数: 116	2020
Anti-SAT: Mitigating SAT Attack on Logic Locking	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Xie, Yang; Srivastava, Ankur	
被引用次数: 149	2019
Reducing Timing Side-Channel Information Leakage Using 3D Integration	
IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING	
Bao, Chongxi; Srivastava, Ankur	

被引用次数: 9	2019
In Situ Stochastic Training of MTJ Crossbars With Machine Learning Algorithms	
ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS	
Mondal, Ankit; Srivastava, Ankur	
被引用次数: 8	2019
A Combined Optimization-Theoretic and Side-Channel Approach for Attacking Strong Physical Unclonable Functions	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Liu, Yuntao; Xie, Yang; Bao, Chongxi; Srivastava, Ankur	
被引用次数: 18	2018
Security-Aware 2.5D Integrated Circuit Design Flow Against Hardware IP Piracy	
COMPUTER	
Xie, Yang; Bao, Chongxi; Srivastava, Ankur	
被引用次数: 29	2017
Low-Power Clock Tree Synthesis for 3D-ICs	
ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS	
Lu, Tiantao; Srivastava, Ankur	
被引用次数: 2	2017
TSV-Based 3-D ICs: Design Methods and Tools	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Lu, Tiantao; Serafy, Caleb; Yang, Zhiyuan; Samal, Sandeep Kumar; Lim, Sung Kyu; Srivastava, Ankur	
被引用次数: 39	2017
On Reverse Engineering-Based Hardware Trojan Detection	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Bao, Chongxi; Forte, Domenic; Srivastava, Ankur	
被引用次数: 115	2016
Thermoelectric Codesign of 3-D CPUs and Embedded Microfluidic Pin-Fin Heatsinks	
IEEE DESIGN & TEST	
Serafy, Caleb; Yang, Zhiyuan; Srivastava, Ankur; Hu, Yuanchen; Joshi, Yogendra	
被引用次数: 8	2016
Unlocking the True Potential of 3-D CPUs With Microfluidic Cooling	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Serafy, Caleb; Bar-Cohen, Avram; Srivastava, Ankur; Yeung, Donald	
被引用次数: 15	2016
Modeling and Layout Optimization for Tapered TSVs	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Lu, Tiantao; Srivastava, Ankur	

被引用次数: 13	2015
Temperature Tracking: Toward Robust Run-Time Detection of Hardware Trojans	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Bao, Chongxi; Forte, Domenic; Srivastava, Ankur	
被引用次数: 72	2015
Optimized Micro-Channel Design for Stacked 3-D-ICs	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Shi, Bing; Srivastava, Ankur	
被引用次数: 22	2014
A geometric approach to chip-scale TSV shield placement for the reduction of TSV coupling in 3D-ICs	
INTEGRATION-THE VLSI JOURNAL	
Serafy, Caleb; Shi, Bing; Srivastava, Ankur	
被引用次数: 9	2014
Improving the Quality of Delay-Based PUFs via Optical Proximity Correction	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Forte, Domenic; Srivastava, Ankur	
被引用次数: 2	2013
Resource-Aware Architectures for Adaptive Particle Filter Based Visual Target Tracking	
ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS	
Forte, Domenic; Srivastava, Ankur	
被引用次数: 0	2013
Dynamic Thermal Management Under Soft Thermal Constraints	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Shi, Bing; Zhang, Yufu; Srivastava, Ankur	
被引用次数: 14	2013
Thermal-Aware Sensor Scheduling for Distributed Estimation	
ACM TRANSACTIONS ON SENSOR NETWORKS	
Forte, Domenic; Srivastava, Ankur	
被引用次数: 9	2013
Accelerating Gate Sizing Using Graphics Processing Units	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Shi, Bing; Zhang, Yufu; Srivastava, Ankur	
被引用次数: 1	2012
Accurate Temperature Estimation Using Noisy Thermal Sensors for Gaussian and Non-Gaussian Cases	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Zhang, Yufu; Srivastava, Ankur	

被引用次数: 39

2011

Unified datacenter power management considering on-chip and air temperature constraints

SUSTAINABLE COMPUTING-INFORMATICS & SYSTEMS

Shi, Bing; Srivastava, Ankur

被引用次数: 5

2011

On-Chip Sensor-Driven Efficient Thermal Profile Estimation Algorithms

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Zhang, Yufu; Srivastava, Ankur; Zahran, Mohamed

被引用次数: 6

2010

Variability-driven formulation for simultaneous gate sizing and postsilicon tunability allocation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Khandelwal, Vishal; Srivastava, Ankur

被引用次数: 12

2008

Variability driven gate sizing for binning yield optimization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Davoodi, Azadeh; Srivastava, Ankur

被引用次数: 21

2008

Algorithmic and architectural optimizations for computationally efficient particle filtering

IEEE TRANSACTIONS ON IMAGE PROCESSING

Sankaranarayanan, Aswin C.; Srivastava, Ankur; Chellappa, Rama

被引用次数: 49

2008

Active mode leakage reduction using fine-grained forward body biasing strategy

INTEGRATION-THE VLSI JOURNAL

Khandelwal, Vishal; Srivastava, Ankur

被引用次数: 7

2007

Leakage control through fine-grained placement and sizing of sleep transistors

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Khandelwal, Vishal; Srivastava, Ankur

被引用次数: 22

2007

A quadratic modeling-based framework for accurate statistical timing analysis considering correlations

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Khandelwal, Vishal; Srivastava, Ankur

被引用次数: 11

2007

Electron beam and optical proximity effect reduction for nanolithography: New results

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Peckerar, Martin; Sander, David; Srivastava, Ankur; Foli, Adakou; Vishkin, Uzi	2007
被引用次数: 15	
A statistical methodology for wire-length prediction	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Wong, Jennifer L.; Davoodi, Azadeh; Khandelwal, Vishal; Srivastava, Ankur; Potkonjak, Miodrag	
被引用次数: 6	2006
Effective techniques for the generalized low-power binding problem	
ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS	
Davoodi, A; Srivastava, A	
被引用次数: 6	2006
Probabilistic evaluation of solutions in variability-driven optimization	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Davoodi, Azadeh; Khandelwal, Vishal; Srivastava, Ankur	
被引用次数: 0	2006
On effective slack management in postscheduling phase	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Srivastava, A; Memik, SO; Choi, BK; Sarrafzadeh, M	
被引用次数: 3	2005
Voltage scheduling under unpredictabilities: A risk management paradigm	
ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS	
Davoodi, A; Srivastava, A	
被引用次数: 0	2005
Simultaneous Vt selection and assignment for leakage optimization	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Khandelwal, V; Davoodi, A; Srivastava, A	
被引用次数: 12	2005
Empirical models for net-length probability distribution and applications	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Davoodi, A; Khandelwal, V; Srivastava, A	
被引用次数: 4	2004
Timing driven gate duplication	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Srivastava, A; Kastner, R; Chen, CH; Sarrafzadeh, M	
被引用次数: 11	2004
Predictability in RT-Level designs	
JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS	
Srivastava, A; Kursun, E; Sarrafzadeh, M	

被引用次数: 2	2002
Optimal energy aware clustering in sensor networks	
SENSORS	
Ghiasi, S; Srivastava, A; Yang, XJ; Sarrafzadeh, M	
被引用次数: 190	2002
Budget management with applications	
ALGORITHMICA	
Chen, CH; Bozorgzadeh, E; Srivastava, A; Sarrafzadeh, M	
被引用次数: 16	2002
Activity-driven clock design	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Farrahi, AH; Chen, CH; Srivastava, A; Téllez, G; Sarrafzadeh, M	
被引用次数: 73	2001
On gate level power optimization using dual-supply voltages	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Chen, CH; Srivastava, A; Sarrafzadeh, M	
被引用次数: 71	2001
Timing driven gate duplication in technology independent phase	
IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES	
Srivastava, A; Chen, CH; Sarrafzadeh, M	
被引用次数: 0	2001
On the complexity of gate duplication	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Srivastava, A; Kastner, R; Sarrafzadeh, M	
被引用次数: 1	2001
Trace Logic Locking: Improving the Parametric Space of Logic Locking	
TCAD	
Michael Zuzak; Yuntao Liu; Ankur Srivastava	
被引用次数: 0	
Evaluating the Security of Logic-Locked Probabilistic Circuits	
TCAD	
Michael Zuzak; Ankit Mondal; Ankur Srivastava	
被引用次数: 0	
TSV Replacement and Shield Insertion for TSV-TSV Coupling Reduction in 3-D Global Placement	
TCAD	
Caleb Serafy; Ankur Srivastava	

被引用次数: 0

On Reverse Engineering-Based Hardware Trojan Detection

TCAD

Chongxi Bao; Domenic Forte; Ankur Srivastava

被引用次数: 0

Temperature Tracking: Toward Robust Run-Time Detection of Hardware Trojans

TCAD

Chongxi Bao; Domenic Forte; Ankur Srivastava

被引用次数: 0

Security-Aware Resource Binding to Enhance Logic Obfuscation

TCAD

Michael Zuzak; Yuntao Liu; Ankur Srivastava

被引用次数: 0

Keynote: A Disquisition on Logic Locking

TCAD

Abhishek Chakraborty; Nithyashankari Gummidipoondi Jayasankaran; Yuntao Liu; Jeyavijayan Rajendran; Ozgur Sinanoglu; Ankur Srivastava; Yang Xie; Muhammad Yasin; Michael Zuzak

被引用次数: 0

Reducing Timing Side-Channel Information Leakage Using 3D Integration

TDSC

Chongxi Bao; Ankur Srivastava

被引用次数: 0

Anti-SAT: Mitigating SAT Attack on Logic Locking

TCAD

Yang Xie; Ankur Srivastava

被引用次数: 0

Evaluating the Security of Delay-Locked Circuits

TCAD

Abhishek Chakraborty; Yuntao Liu; Ankur Srivastava

被引用次数: 0

TSV-Based 3-D ICs: Design Methods and Tools

TCAD

Tiantao Lu; Caleb Serafy; Zhiyuan Yang; Sandeep Kumar Samal; Sung Kyu Lim; Ankur Srivastava

被引用次数: 0

学术合作



Chongxi Bao

马里兰大学-大学城

合作次数: 5

SUNG KYU LIM

佐治亚理工学院

合作次数: 1

Sandeep Kumar Samal

佐治亚理工学院

合作次数: 1

Zhiyuan Yang

马里兰大学-大学城

合作次数: 1

Domenic Forte

佛罗里达大学

合作次数: 1

还有 6 个合作者